

Laboratory report of Nanjing University of Information Science and Technology

Experiment Title Cloud-Native Automated Deployment Data 2026.5.11
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一、Experimental purpose

The primary goal of this experiment is to introduce students to the core concepts of Cloud Computing through hands-on practice with Platform-as-a-Service (PaaS) and Automation.

二、Experiment Contents

This lab consists of five integrated modules designed to simulate a professional cloud deployment lifecycle.

This experiment uses GitHub, GitHub Pages, and GitHub Actions to complete a cloud-native automated deployment process. First, I forked the original repository to create my own copy of the project. Then, I enabled GitHub Pages and selected GitHub Actions as the deployment source. After that, I edited the index.html file and changed the placeholder name to my own name. Once I committed the change, GitHub Actions automatically started the deployment pipeline. Finally, I opened the generated GitHub Pages URL and verified that the website was successfully deployed online.

三、Detailed steps of the experiment

Step 1: Resource Acquisition

Log in to GitHub.

Navigate to the repository:

<https://github.com/yhldrf/cloud-computing-lab-github>

and click Fork.

The image shows two screenshots from the GitHub interface. The top screenshot is the 'Create a new fork' page for the repository 'cloud-computing-lab-github' by user 'yhldrf'. It includes a search bar, navigation tabs (Code, Issues, Pull requests, Actions, Projects, Security and quality, Insights), and a form to create a fork. The form has fields for 'Owner' (set to 'WentingTang'), 'Repository name' (set to 'cloud-computing-lab-github'), and 'Description' (set to 'fork'). A checkbox 'Copy the main branch only' is checked. A green 'Create fork' button is at the bottom right. The bottom screenshot shows the resulting forked repository page for 'WentingTang / cloud-computing-lab-github'. It displays the repository's public status, a list of files (index.html, .github/workflows), and a 'README' section with an 'Add a README' button. The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, and 0 releases.

Cloud Concept: You are creating a private instance of the project in the cloud ecosystem.

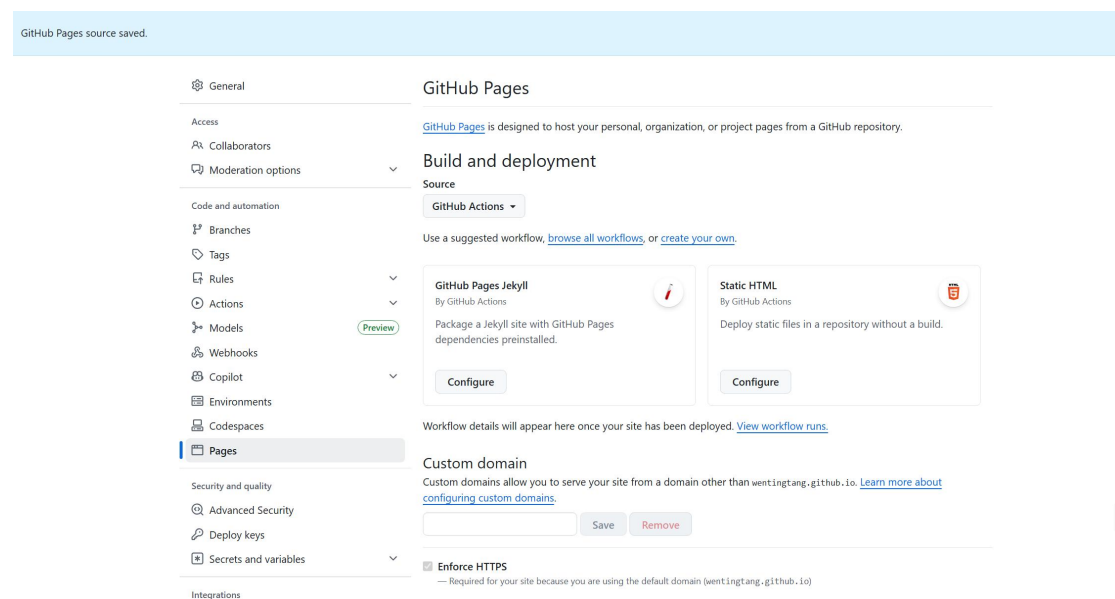
Step 2: Enabling the Hosting Service

Go to Settings > Pages.

Under Build and deployment > Source, select "GitHub Actions".

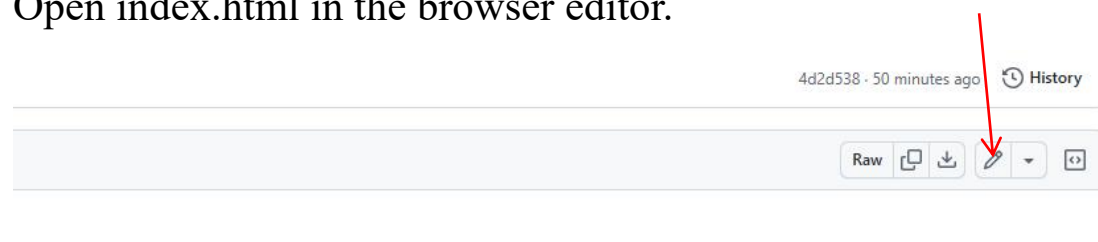
After that, leave this page.

Cloud Concept: You are switching from "Manual Upload" to "Managed Pipeline Service."

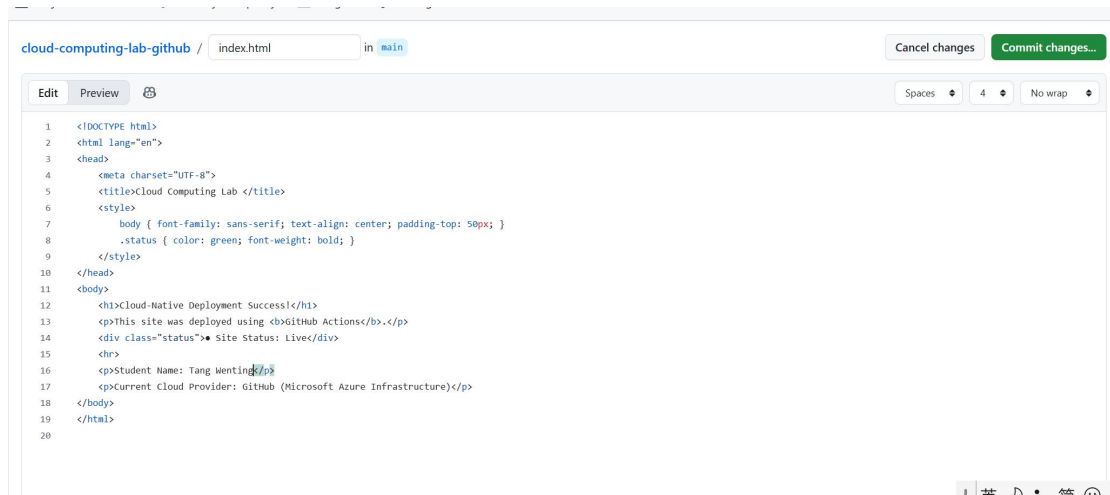


Step 3: Triggering Cloud Automation

Open index.html in the browser editor.



Change [Enter Your Name Here] to your actual name.

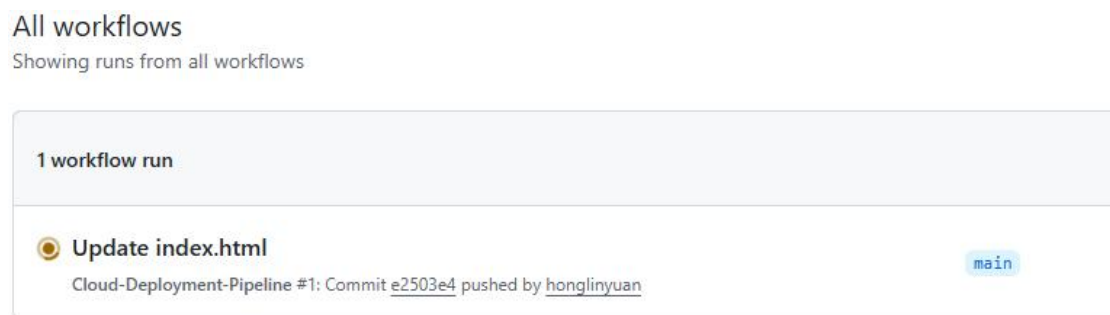


```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Cloud Computing Lab </title>
6   <style>
7     body { font-family: sans-serif; text-align: center; padding-top: 50px; }
8     .status { color: green; font-weight: bold; }
9   </style>
10 </head>
11 <body>
12   <h1>Cloud-Native Deployment Success!</h1>
13   <p>This site was deployed using <b>GitHub Actions</b>.</p>
14   <div class="status"> Site Status: Live</div>
15   <hr>
16   <p>Student Name: Tang Wenting</p>
17   <p>Current Cloud Provider: Github (Microsoft Azure Infrastructure)</p>
18 </body>
19 </html>
20
```

Click Commit Changes.

Immediately click the Actions tab at the top of the page.

Observation: Watch the "Cloud-Deployment-Pipeline" start.



All workflows

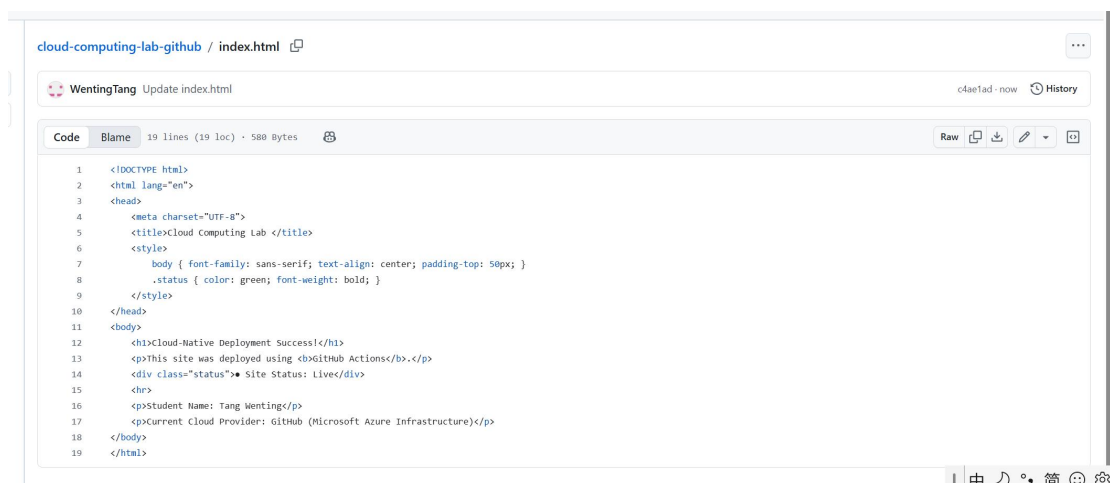
Showing runs from all workflows

1 workflow run

Update index.html

Cloud-Deployment-Pipeline #1: Commit `e2503e4` pushed by `honglin yuan`

Cloud Concept: GitHub is spinning up a Virtual Machine (Ubuntu) to process your request. This is "On-demand Computing."

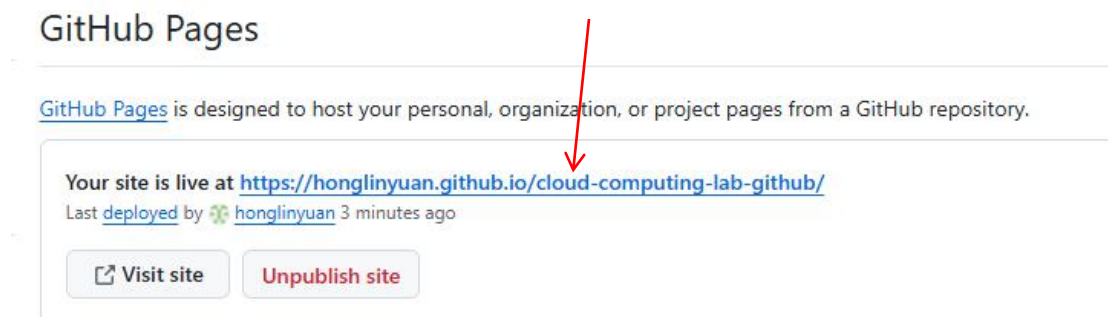


```
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17   <p>Current Cloud Provider: Github (Microsoft Azure Infrastructure)</p>
18 </body>
19 </html>
20
```

WentingTang Update index.html c4ae1ad · 2 minutes ago 4 Commits		
 .github/workflows	Create static.yml	8 hours ago
 index.html	Update index.html	2 minutes ago

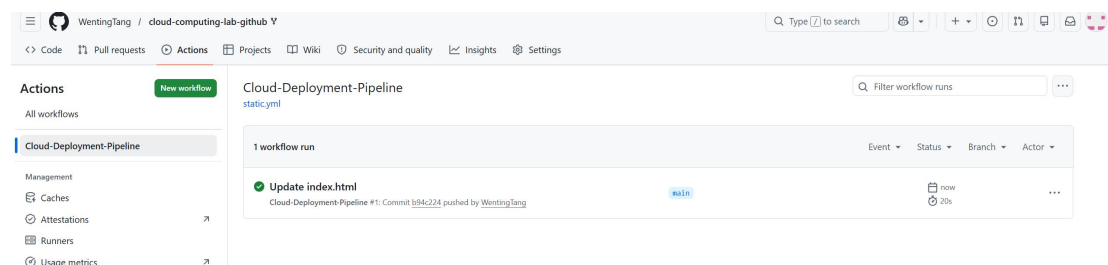
Step 4: Verification & Global Access

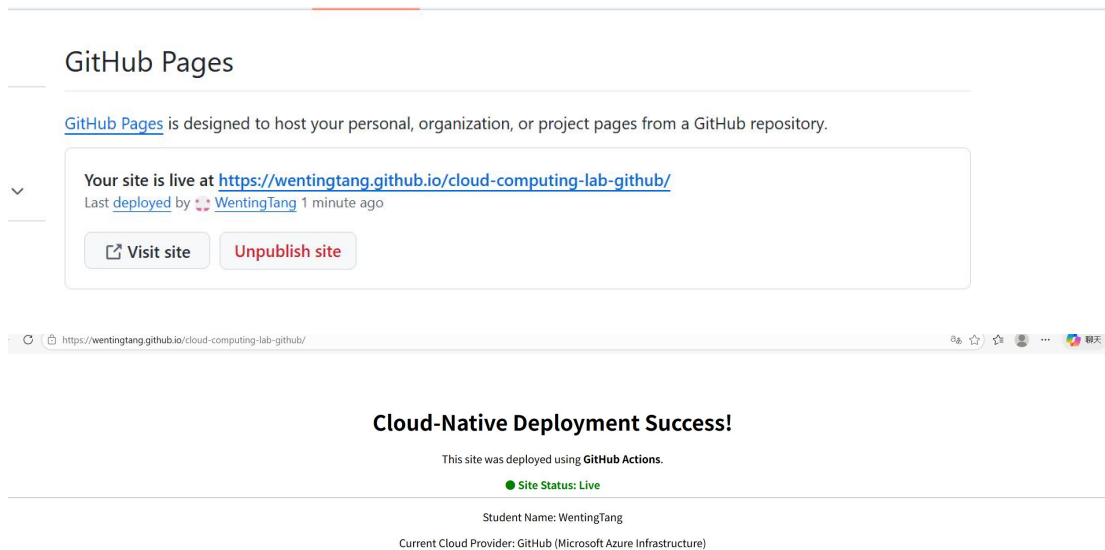
Once the Action finishes (**green check**), go back to Settings > Pages to find your Live URL.



Open the URL on your laptop.

Cloud Concept: The site is deployed via a PaaS (Platform as a Service) model. The cloud provider manages the underlying infrastructure and global delivery, allowing us to focus only on our content.





Step5: Content Personalization & Elements (optional)

Task: Beyond changing your name, modify the following HTML elements to see how the cloud updates different types of content.

✓ Change the Theme Color (CSS Styles)

Find the<style>block at the top and change the background color or font size.

Action:

```
<style>
  body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    text-align: center;
    padding-top: 50px;
    background-color: #f4f4f9; /* Add a light grey background */
  }
  .status { color: #007bff; font-weight: bold; } /* Change status to Blue */
</style>
```

Cloud Concept: You are modifying the User Interface (UI)

settings stored in the cloud.

✓ Multimedia Integration (Images)

Demonstrate how the cloud handles external resources.

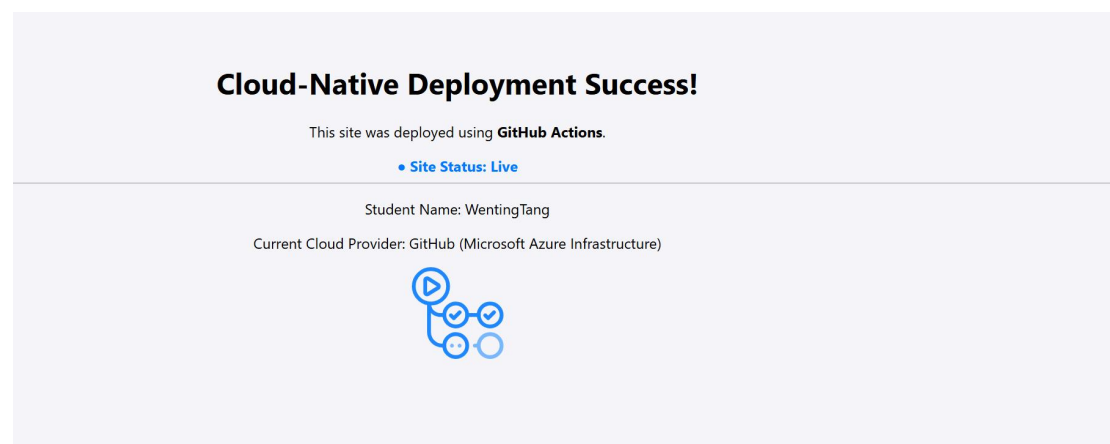
What to add: Antag inside the<body>.

Example:

```

```

Observation: Notice that as soon as you click "Commit," the Actions tab triggers a new build automatically. This is Continuous Deployment (CD).



四、Reflection

1. Where is your website stored?

The website is stored and hosted on GitHub Pages. All website files are saved in the GitHub cloud repository, and GitHub automatically deploys the webpage to the internet through GitHub Actions and GitHub Pages services.

2. Did you have to install any software?

No, I did not need to install any additional software. All operations, including editing

files, committing changes, and deploying the website, were completed directly through the GitHub web interface in the browser.

3. Problems Encountered and Proposed Solutions

During the experiment, I encountered several problems. At first, the GitHub Actions workflow did not run automatically after editing the file. Later, I found that I needed to commit the changes directly to the main branch to trigger the deployment pipeline. Another issue was that the Live URL did not appear immediately because the deployment process had not finished yet. I solved the problem by checking the Actions page, waiting for the workflow to complete successfully, and then refreshing the GitHub Pages settings page.